3-5 Zeros of Polynomial Functions

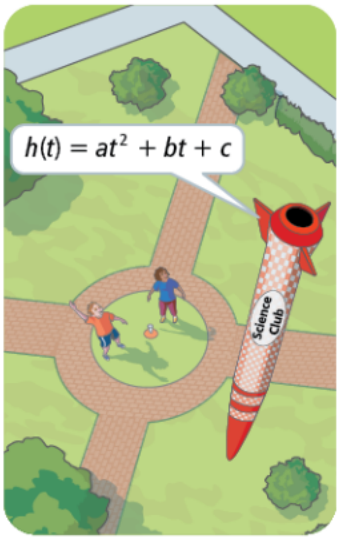
I CAN… model and solve problems using the zeros of a polynomial function.

VOCABULARY

- multiplicity of a zero

MODEL & DISCUSS

Charlie's and Aisha's STEM club built a small rocket and launched it from a field. The rocket fell to the ground 5 s after it launched.



The height h, in feet, of the rocket relative to the ground at time t seconds can be modeled by the function shown.

A. How are the launch and landing times related to the modeling function?

B. What additional information about the rocket launch could you use to construct an accurate model for the rocket's height relative to the ground?

C. **Construct Arguments** Charlie believes that the function $h(t) = -16t^2 + 80t$ models the height of the rocket with respect to time. Do you agree? Explain your reasoning and indicate the domain of this function.

ESSENTIAL QUESTION How are the zeros of a polynomial function related to an equation and graph of the function?

CONCEPTUAL UNDERSTANDING

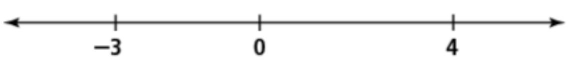
GENERALIZE
Recall using the Zero-Product Property to find the zeros of quadratic polynomials. The zeros of a quadratic function relate to the factors: when $x = a$ is a zero of the function, then $(x - a)$ is a factor of the polynomial. The Zero-Product Property applies to polynomials of any degree.

EXAMPLE 1 Use Zeros to Graph a Polynomial Function

What are the zeros of $f(x) = x(x - 4)(x + 3)$? Graph the function.

A *zero* of a polynomial function is a value for which the function is equal to 0. By the Zero-Product Property, the zeros of the function are -3, 0, and 4.

The zeros divide a number line into four intervals.



To see how the graph of the function behaves on each interval, look at the sign of each factor on each interval, and the sign of the product.

Interval	Sign			
	x	$x - 4$	$x + 3$	Product
$x < -3$	–	–	–	–
$-3 < x < 0$	–	–	+	+
$0 < x < 4$	+	–	+	–
$x > 4$	+	+	+	+

The last column shows the sign of the product of the three factors.

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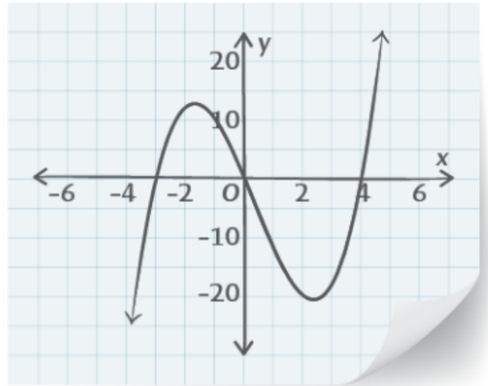
STUDY TIP
The zeros do not tell you how a function behaves on an interval, just that a function may change between positive or negative at a zero.

EXAMPLE 1 CONTINUED

Sketch the graph. Draw a continuous curve that passes through each zero on the x-axis, and is below the x-axis when the function is negative, and above the x-axis when the function is positive.



Try It! **1.** Factor each function. Then use the zeros to sketch its graph.

a. $f(x) = 4x^3 + 4x^2 - 24x$ **b.** $g(x) = x^4 - 81$

EXAMPLE 2 Understand How a Multiple Zero Can Affect a Graph

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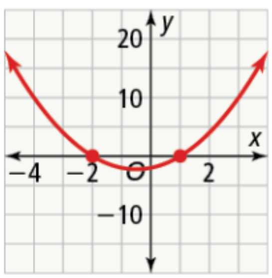
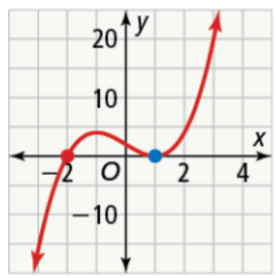
How does a multiple zero affect the graph of a polynomial function?

The **multiplicity of a zero** of a polynomial function is the number of times its related factor appears in the factored form of the polynomial. Notice the behavior of each graph as it approaches the x-axis. What can you conclude about the multiplicity of a zero and its effect on the graph of the function?

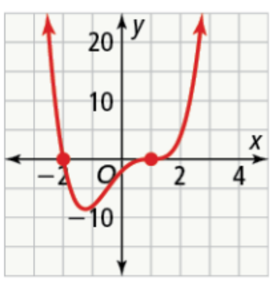
STUDY TIP
Close to a zero, the graph looks like a polynomial function with degree equal to the multiplicity of the zero.



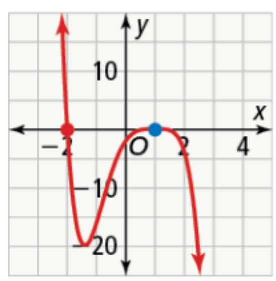
$f(x) = (x - 1)(x + 2)$



$f(x) = (x - 1)^2(x + 2)$



$f(x) = (x - 1)^3(x + 2)$



$f(x) = -(x - 1)^4(x + 2)$

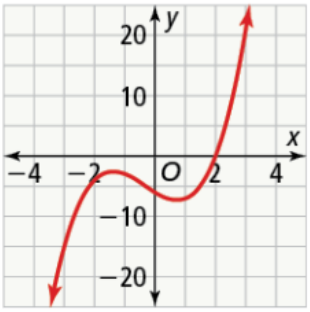
When the multiplicity of a zero is odd, the function crosses the x-axis. When the multiplicity of a zero is even, the graph has a turning point at the x-axis.

Try It! **2.** Describe the behavior of the graph of the function at each of its zeros.

a. $f(x) = x(x + 4)(x - 1)^4$ **b.** $f(x) = (x^2 + 9)(x - 1)^5(x + 2)^2$

EXAMPLE 3 Find Real and Complex Zeros

What are all the real and complex zeros of the polynomial function shown in the graph?



$f(x) = x^3 + x^2 - 3x - 6$

Step 1 Use the graph to determine one of the zeros of the polynomial. The function appears to cross the x-axis at $x = 2$.

Confirm that 2 is a zero of the function.

$$f(2) = (2)^3 + (2)^2 - 3(2) - 6$$
$$= 0$$

So 2 is a zero of the function. By the Factor Theorem, $x - 2$ is a factor of the related polynomial.

Step 2 Use synthetic division to factor the polynomial.

$$\begin{array}{r|rrrr} 2 & 1 & 1 & -3 & -6 \\ & & 2 & 6 & 6 \\ \hline & 1 & 3 & 3 & 0 \end{array}$$

So $f(x) = (x - 2)(x^2 + 3x + 3)$.

Step 3 Use the Quadratic Formula to find the remaining zeros.

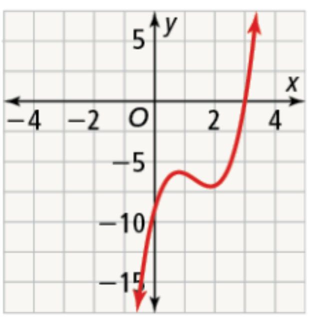
$$x = \frac{-3 \pm \sqrt{3^2 - 4(1)(3)}}{2(1)}$$
$$= -\frac{3}{2} \pm \frac{\sqrt{3}}{2}i$$

The zeros of the polynomial function f are 2, $-\frac{3}{2} + \frac{\sqrt{3}}{2}i$, and $-\frac{3}{2} - \frac{\sqrt{3}}{2}i$.

> **LOOK FOR RELATIONSHIPS**
> These statements are equivalent:
> - The function's graph crosses the x-axis at 2.
> - The x-intercept of the graph is 2.
> - $f(2) = 0$
> - 2 is a zero of the function.
> - $x - 2$ is a factor of the polynomial.

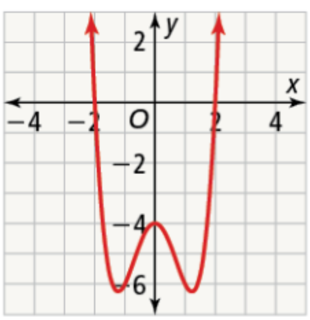
Try It! **3.** What are all the real and complex zeros of the polynomial function shown in the graph?

a.



$f(x) = 2x^3 - 8x^2 + 9x - 9$

b.



$f(x) = x^4 - 3x^2 - 4$

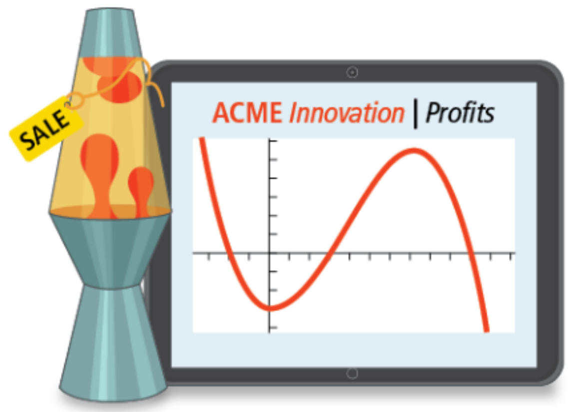
APPLICATION

EXAMPLE 4 Find the Polynomial Model

Acme Innovations makes and sells lamps. Their profit P, in hundreds of dollars earned, is a function of the number of lamps sold x, in thousands.

From historical data, they know that their company's profit is modeled by the function shown.

The function has zeros at −2, 3, and 10. When Acme Innovations sells 2,000 lamps, they lose \$3,200. Find P in standard form and interpret the key features for this situation.



Formulate

A profit for the company corresponds to the portions of the graph that lie in the first quadrant. Use the zeros of the function to determine the factors of P.

By the Factor Theorem you can determine the factors of the related polynomial.

Zero of p	*Factor of p(x)*
-2	$x+2$
3	$x-3$
10	$x-10$

By the end behavior, the polynomial's leading coefficient is negative, and the function's degree is odd. Since there are three distinct zeros, the least degree the polynomial could have is 3.

Compute

REASON
Before multiplying to find the standard form of the function, what could you do to determine the multiplicity of the factors or a stretch factor?

Multiply these factors and −1 to find the standard form of the polynomial.

$$\begin{aligned}-(x+2)(x-3)(x-10) &= (-x-2)(x^2-13x+30)\\ &= -x^3+13x^2-30x-2x^2+26x-60\\ &= -x^3+11x^2-4x-60\end{aligned}$$

Find $P(2)$ to see if polynomial goes through $(2, -32)$.

$$\begin{aligned}P(2) &= -(2)^3+11(2)^2-4(2)-60\\ &= -8+44-8-60\\ &= -32\end{aligned}$$

So $P(x) = -x^3+11x^2-4x-60$.

Interpret

From the y-intercept, Acme Innovations spends \$6,000 to set up production before they make the first lamp. When they make and sell 3,000 or 10,000 lamps, they break even. They make a profit when they make and sell between those values. Using technology, you can find they earn the most when they make and sell 7,147 lamps for a profit of \$10,822.

Try It! 4. Due to a decrease in the cost of materials, the profit function for Acme Innovations has changed to $Q(x) = -x^3+10x^2+13x-22$. How many lamps should they make in order to make a profit?

EXAMPLE 5 Solve Polynomial Equations

What are the solutions of $2x^3 + 5x^2 - 3x = 3x^3 + 8x^2 + 1$?

Rewrite the equation in the form $P(x) = 0$.

$$2x^3 + 5x^2 - 3x = 3x^3 + 8x^2 + 1$$
$$x^3 + 3x^2 + 3x + 1 = 0$$

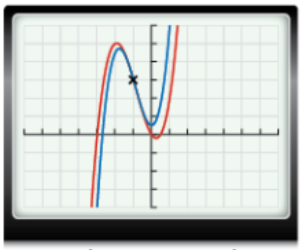
Combine like terms on one side of the equation.

VOCABULARY
A *polynomial equation* is an equation that can be written in the form $P(x) = 0$, where $P(x)$ is a polynomial.

The roots are the zeros of the function $P(x) = x^3 + 3x^2 + 3x + 1$.

$$(x + 1)^3 = 0$$
$$x + 1 = 0$$
$$x = -1$$

To check, write each side of the equation as a separate polynomial and graph. Use the INTERSECT feature to confirm that the graphs intersect at $x = -1$.



x scale: 1   y scale: 2

Try It! 5. What is the solution of the equation?

a. $x^3 - 7x + 6 = x^3 + 5x^2 - 2x - 24$ b. $x^4 + 2x^2 = -x^3 - 2x$

EXAMPLE 6 Solve a Polynomial Inequality by Graphing

HAVE A GROWTH MINDSET
How can you take on challenges with positivity?

What are the solutions of $x^3 - 16x < 0$?

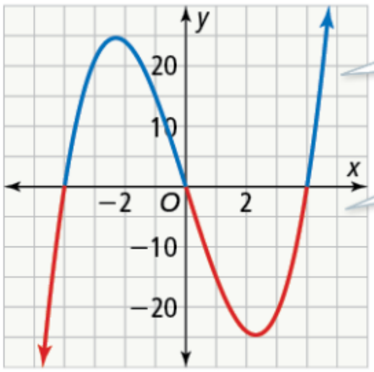
The solutions are all values of x that make the inequality true. The polynomial $x^3 - 16x$ defines a polynomial function $P(x) = x^3 - 16x$.

Factor to find the zeros of the function.

$$x^3 - 16x = 0$$
$$x(x^2 - 16) = 0$$
$$x(x - 4)(x + 4) = 0$$

By the Zero-Product Property, the zeros of P are -4, 0, and 4.

Sketch the function, and use the graph to determine where $P(x) < 0$.



The blue portions show where $P(x) > 0$.

The red portions show where $P(x) < 0$.

The solutions of the inequality $x^3 - 16x < 0$ are all real numbers such that $x < -4$ or $0 < x < 4$.

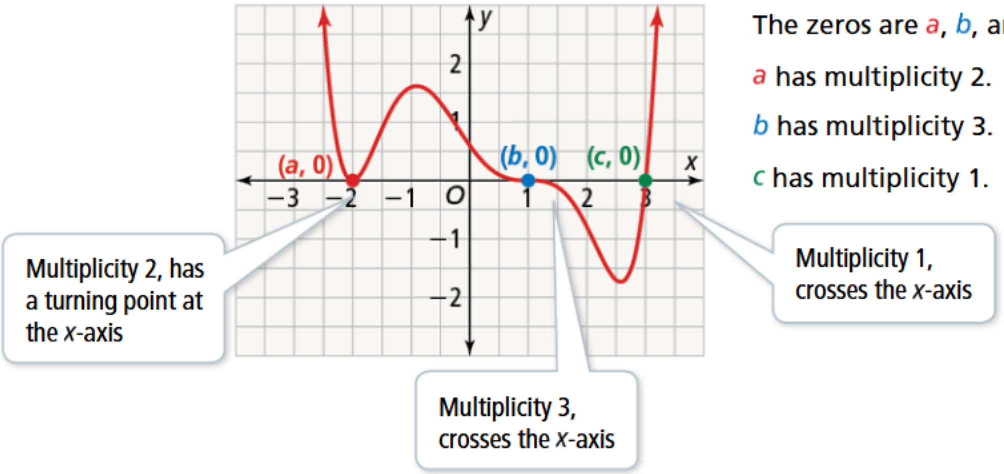
Try It! 6. What are the solutions of the inequality?

a. $2x^3 + 12x^2 + 12x < 0$ b. $(x^2 - 1)(x^2 - x - 6) > 0$

CONCEPT SUMMARY Polynomial Identitiess

FUNCTION $f(x) = (x - a)^2(x - b)^3(x - c)$

GRAPH



The zeros are a, b, and c.

a has multiplicity 2.

b has multiplicity 3.

c has multiplicity 1.

Do You UNDERSTAND?

1. **ESSENTIAL QUESTION** How are the zeros of a polynomial function related to the equation and graph of a function?

2. **Error Analysis** In order to identify the zeros of the function, a student factored the cubic function $f(x) = x^3 - 3x^2 - 10x$ as follows:

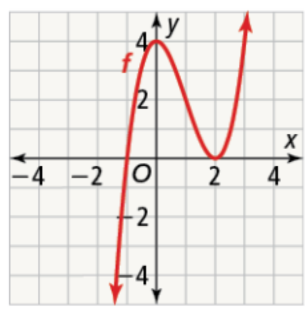
   $f(x) = x^3 - 3x^2 - 10x$
   $= x(x^2 - 3x - 10)$
   $= x(x - 5)(x + 5)$
   $x = 0, x = -5, x = 5$

   Describe and correct the error the student made. **MP.3**

3. **Make Sense and Persevere** Explain how you can determine that the function $f(x) = x^3 + 3x^2 + 4x + 2$ has both real and complex zeros **MP.1**

Do You KNOW HOW?

4. If the graph of the function f has a multiple zero at $x = 2$, what is a possible exponent of the factor $x - 2$? Justify your reasoning.



5. Energy Solutions manufactures LED light bulbs. The profit p, in thousands of dollars earned, is a function of the number of bulbs sold, x, in ten thousands. Profit is modeled by the function $-x^3 + 9x^2 - 11x - 21$.

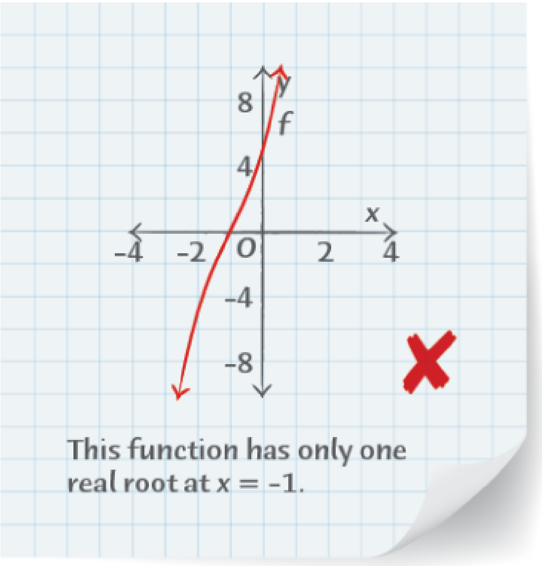
   For what number of bulbs manufactured does the company make a profit?

PRACTICE & PROBLEM SOLVING

UNDERSTAND

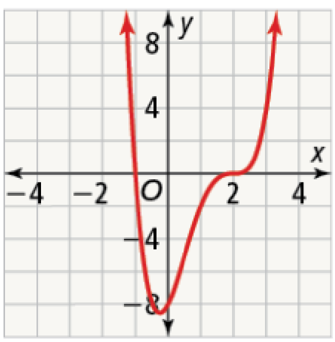
6. **Reason** If you use zeros to sketch the graph of a polynomial function, how can you verify that your graph is correct?

7. **Error Analysis** Describe and resolve two errors that Tonya may have made in finding all the roots of the polynomial function, $f(x) = x^3 + 3x^2 + 7x + 5$.



8. **Higher Order Thinking** How could you use your graphing calculator to determine that $f(x) = (x + 2)(x + 6)(x - 1)$ is not the correct factorization of $f(x) = x^3 + 7x^2 + 16x + 12$? Explain.

9. **Generalize** How can you determine that the polynomial function shown does not have any zeros with even multiplicity? Explain.



10. **Use Structure** Factor the polynomial $x^4 - 16$. How many real zeros does the function $g(x) = x^4 - 16$ have?

11. At what points do the graphs of $f(x) = x^3 - 2x^2 - 16x + 20$ and $g(x) = -12$ intersect?

PRACTICE

Sketch the graph of the function by finding the zeros. SEE EXAMPLE 1

12. $f(x) = 3x^3 - 9x^2 - 12x$

13. $g(x) = (x + 3)(x - 1)(x - 4)$

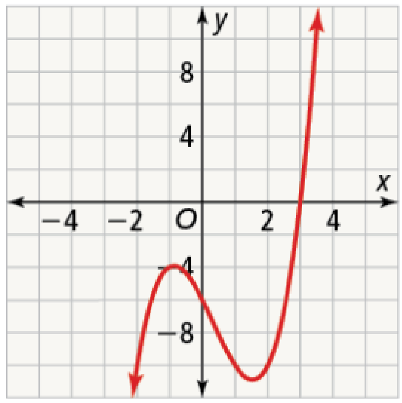
Find the zeros of the function, and describe the behavior of the graph at each zero. SEE EXAMPLE 2

14. $f(x) = x^3 - 8x^2 + 16x$

15. $g(x) = x^3 - x^2 - 25x + 25$

16. $f(x) = 9x^4 - 40x^2 + 16$

17. What are all the real and complex zeros of the polynomial function shown in the graph? SEE EXAMPLE 3



$f(x) = x^3 - x^2 - 4x - 6$

18. Waterworks is a company that manufactures and sells paddleboards. Their profit P, in hundreds of dollars earned, is a function of the number of paddleboards sold x, measured in thousands. Profit is modeled by the function $P(x) = -3x^3 + 48x^2 - 144x$. What do the zeros of the function tell you about the number of paddleboards that Waterworks should produce? SEE EXAMPLE 4

What are the solution(s) of the equation? SEE EXAMPLE 5

19. $-3x^3 - x^2 + 54x - 40 = 2x^2 + 6x + 20$

20. $2x^3 + 3x^2 - 36 = x^3 - x^2 + 9x$

21. $-5x^4 + 4x^2 - 12x = -6x^4 + 3x^3$

What are the solutions of the inequality? SEE EXAMPLE 6

22. $x^3 - 9x > 0$

23. $0 > 4x^3 + 8x^2 - x - 2$

24. $64x^2 > -4x^3 - x - 16$

PRACTICE & PROBLEM SOLVING

APPLY

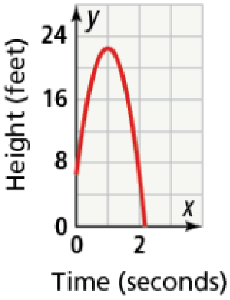
25. **Make Sense and Persevere** A firework is launched vertically into the air. Its height in meters is given by the function shown, where t is measured in seconds.

a. What is a reasonable domain of the function?

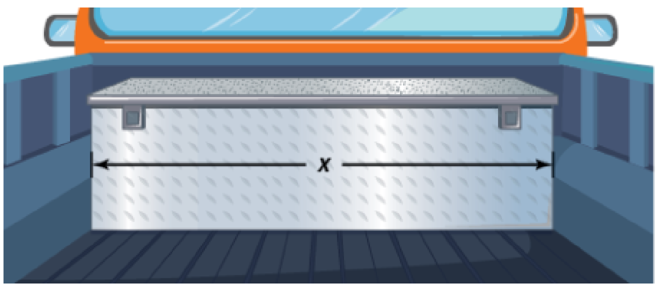
b. What are the zeros of the function? Explain what they represent in this situation.

c. Use technology to find the vertex. What does it represent in this situation?

26. The height of a baseball thrown in the air can be modeled by the function $h(t) = -16t^2 + 32t + 6.5$, where $h(t)$ represents the height in feet of the baseball after t seconds. Explain why the graph of this function only shows one zero.



27. **Model With Mathematics** The height of a rectangular storage box is less than both its length and width. The function $f(x) = x^3 + 2x^2 - 3x$ represents the volume of the rectangular box, where x represents the width of the box, in feet.



a. Find the factored form of $f(x)$.

b. Find the zeros of the function.

c. You know x represents the width of the box. What do the other two factors represent?

d. Find the dimensions of the box when the volume is 10 ft^3.

ASSESSMENT PRACTICE

28. Complete each statement so it means the same as 4 is *a zero of the function*.

The graph of the function crosses the ________ at 4. ____________ is a factor of the polynomial.

29. **SAT/ACT** Without the use of a graphing calculator, determine which of the following functions is the graph of $f(x) = x^3 + x^2 - 4x$.

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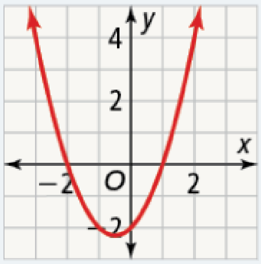

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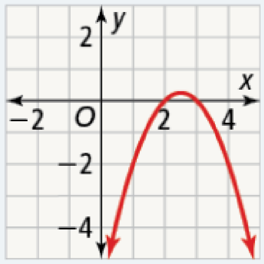

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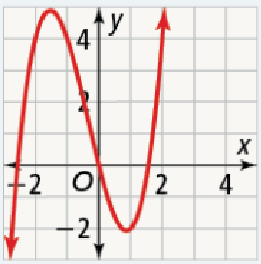

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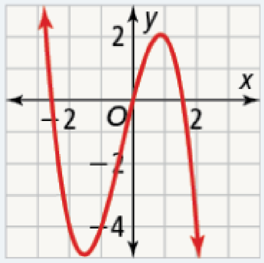

30. **Performance Task** Venetta opened several deli sandwich franchises in 2000. The profit P (in hundreds of dollars) of the franchises in t years (since the franchises opened) can be modeled by the function $P(t) = t^3 + t^2 - 6t$.

Part A Sketch a graph of the function.

Part B Based on the model, during what years did Venetta not make a profit?

Part C If the model is appropriate, predict the amount of profit Venetta will receive from her franchises in 2020.

MATHEMATICAL MODELING IN 3 ACTS

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What Are the Rules?

All games have rules about how to play the game. The rules outline such things as when a ball is in or out, how a player scores points, and how many points a player gets for each winning shot.

If you didn't already know how to play tennis, or some other game, could you figure out what the rules were just by watching? What clues would help you understand the game? Think about this during the Mathematical Modeling in 3-Acts lesson.

ACT 1 Identify the Problem

1. What is the first question that comes to mind after watching the video?
2. Write down the main question you will answer about what you saw in the video.
3. Make an initial conjecture that answers this main question.
4. Explain how you arrived at your conjecture.
5. What information will be useful to know to answer the main question? How can you get it? How will you use that information?

ACT 2 Develop a Model

6. Use the math that you have learned in this Topic to refine your conjecture.

ACT 3 Interpret the Results

7. Did your refined conjecture match the actual answer exactly? If not, what might explain the difference?